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Blown Wing Aerodynamic Coefficient Predictions Using Traditional Machine Learning and Data Science Approaches

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Abstract

The rich body of literature on blown wings was leveraged to develop several machine learning and data science algorithms to predict the lift coefficient and pitching moment coefficient for any blown wing configuration. The predictions were made based on over 20 different numerical and categorical input variables such as airfoil thickness, airfoil chord, airfoil camber, angle of attack, Mach number, aspect ratio, etc., along with the engine geometry, engine location, the thrust coefficient, and more. The Gaussian progression model and neural network provided lift and pitching moment coefficient predictions within 5% error. These predictions from the machine learning models are intended to aid preliminary aircraft design or wind tunnel testing. Additional numerical analysis techniques, including sequential forward selection (SFS) were performed to study the feature space and determine the sensitivity of each input variable on lift and moment coefficient. Most models agreed that the thrust coefficient and flap angle were the two most important features for predicting these aerodynamic coefficients.

I. Nomenclature

A	Propulsion System Frontal Area
AR	Aspect Ratio
b	Wingspan
C	Camber, *%*
C_L	Lift Coefficient
$C_{L,max}$	Maximum Lift Coefficient
C_m	Pitching Moment Coefficient
C_T	Thrust Coefficient
h	Fan Height, *ft.*
L	Lift Force, *lb.*
n	Number of Motors
q_∞	Freestream Dynamic pressure, *psi*
R	Fan Area Ratio
Re	Reynolds Number
S	Wing Planform Area, $ft.^2$
$\bar{S}$	Average Value of Data
S_a	Individual Data Point
$S_{a_{norm}}$	Normalized Data Point
S_{max}	Maximum Value of Data
S_{min}	Minimum Value of Data
T	Thrust Force, *lb.*
t/c	Thickness to Chord Ratio
T_ρ	Thrust per Pressure, $ft.^2$
V	Freestream velocity, *ft/s*
V_{stall}	Stall Speed, *ft/s*
W	Aircraft Weight, *lb.*
WC	Wing Coverage
x	Fan Chord Location, *ft.*
α	Angle of Attack, *degree*
δ	Flap Angle, *degree*
ρ	Air Density, $slug/ft^3$
υ	Volume, ft^3

II. Acronyms

DEP	Distributed Electric Propulsion
EBF	Externally Blown Flap
MRMR	Minimum Redundancy Maximum Relevance
NASA	National Aeronautics and Space Administration
NCA	Neighborhood Component Analysis
NN	Neural Network
RMSE	Root Mean Square Error
SFS	Sequential Forward Selection
STOL	Short Takeoff and Landing
SVM	Support Vector Machines
UAM	Urban Air Mobility
USB	Upper Surface Blowing
VTOL	Vertical Takeoff and Landin

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III. Introduction

The aerospace industry today has taken an increasing interest in urban air mobility (UAM) vehicles as an alternative transportation method over short distances in urban areas. Based on a 2018 study conducted by NASA, air metro and last-mile parcel can become profitable markets with large-scale use by 2030. [1] Given the future potential, different system level and component level technologies are being developed by multitude of organizations for this type of use.

Promoted by NASA's strategic implementation plan, the convergence of electric power and distributed propulsion has become a strong contender in future aircraft designs. [1] However, blending electric distributed propulsion and wings creates complex interactions between wing aerodynamics and the propulsion system, the performance of which is difficult to predict without wind tunnel testing or high fidelity CFD which is time consuming and expensive. Using simple yet accurate models, this challenge can be overcome to help design aircraft with improved performance and efficiency in multiple flight regimes, especially for short-haul missions. This is true in large part to the coupling of propulsion and aircraft aerodynamics to work together for reducing drag and improving lift. [2] Apart from improved performance, stability is also a critical requirement for successful UAM vehicles in emerging markets. The purpose of the research presented here is to employ machine learning algorithms to reduce the gap between research and application of distributed electric propulsion (DEP) in UAM vehicles, especially since the use of DEP in an aircraft can improve its efficiency by approximately 4.8 times that of an avgas vehicle. [3] Specific attention is given to the use of blown wings which will aid in short takeoff and landing (STOL) vehicle design.

STOL aircraft typically include aerodynamics that focus on minimizing the stall speed to reduce the takeoff and landing ground roll. Figure 1 below shows a variety of parameters that affect the landing distance of an arbitrary aircraft. [4] The parameter affecting landing distance the greatest is the stall speed. This is also true for takeoff distance where the takeoff speed (which determines ground roll distance) is typically defined as 20% greater than the stall speed. [5] If an aircraft can land or take off at an extremely slow speed, the ground roll will be minimized which is a highly desired constraint for UAM vehicles.

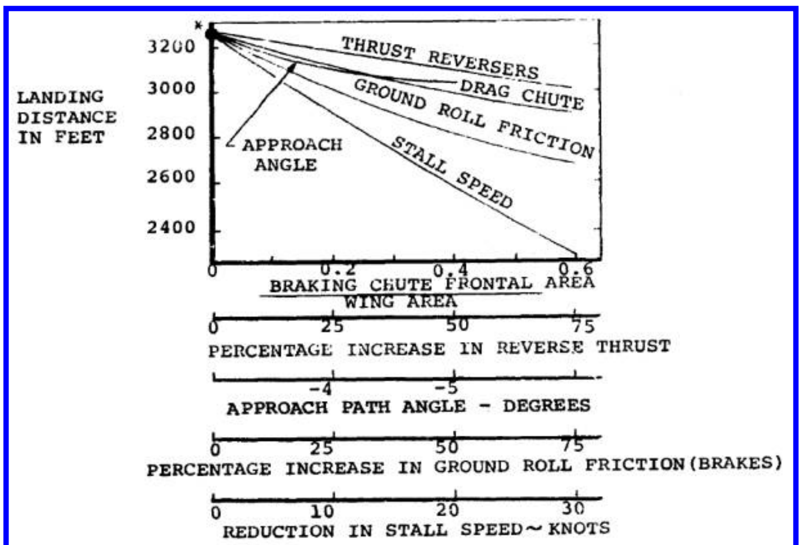

Figure 1. Parameters affecting landing distancing for an example aircraft. [4]

To take off in a short distance, the force of lift must be sufficiently greater than the aircraft weight. Several parameters can be adjusted to increase the force of lift as shown in the following equation

$$L = \frac{1}{2}\rho V^2 S C_L \tag{1}$$

where L is aerodynamic lift, ρ is air density, S is the wing planform area, V^2 is the freestream velocity, and C_L is the coefficient of lift. In stall conditions, the equation can be arranged as follows

$$V_{stall} = \sqrt{\frac{1}{\frac{1}{2}\rho C_{L,max}}\frac{W}{S}} \tag{2}$$

where W is the aircraft weight equivalent to the lift generated and $C_{L,max}$ is the maximum lift coefficient that can be achieved. With fixed air density at a given altitude and an aircraft with fixed weight and wing area, the only way to reduce V_{stall} in this equation is by increasing $C_{L,max}$. Vertical takeoff and landing (VTOL) aircraft overcome the issue of increasing $C_{L,max}$ by increasing the thrust to weight ratio and directing a larger component of thrust to contribute to lift. This reduces the amount of aerodynamic lift required to overcome the weight. STOL aircraft can implement a combination of vertical thrust and aerodynamic lift for takeoff and landing. A major advantage of STOL aircraft is the greatly reduced thrust requirement for takeoff compared to VTOL.

There are a multitude of options for high lift devices such as flaps, slats, and boundary layer control. Each device has several shapes and can be powered or unpowered. The controlled flow around the propulsion sources can be used to artificially increase wing circulation by blowing and thereby increasing $C_{L,max}$. Figure 2 shows several blown wing configurations with different combinations of jet engine-wing proximity and flap settings. Figure 3 shows the corresponding drag polar for each of these configurations. [5] If $C_{L,max}$ is increased from, say, 1.5 to 6.0 due to an externally blown flap (EBF) or upper surface blowing (USB), the stall speed is reduced by a factor of two. An EBF generates this additional lift by blowing the air under the wing and around slotted flaps that extend well beyond the trailing edge. These flaps direct some of the propulsive flow downward, like vectored thrust, to generate more lift. In addition, the slotted flaps allow air to flow to the upper surface for boundary layer control that increases the lift generated by the flaps. The USB configuration, on the other hand, relies predominantly on the Coanda effect to redirect the momentum of the jet exhaust in the downward direction to augment lift. The exhaust flow travels across the wing and continues to adhere to the surface of the wing as it travels down a deflected flap. These designs, while serving as a great candidates for generating high amounts of lift and reducing takeoff distance, the complexity and weight associate with such systems greatly reduce the overall aircraft efficiency [6]. Therefore, tradeoff between performance at takeoff and cruise conditions requires a balance in the design choices.

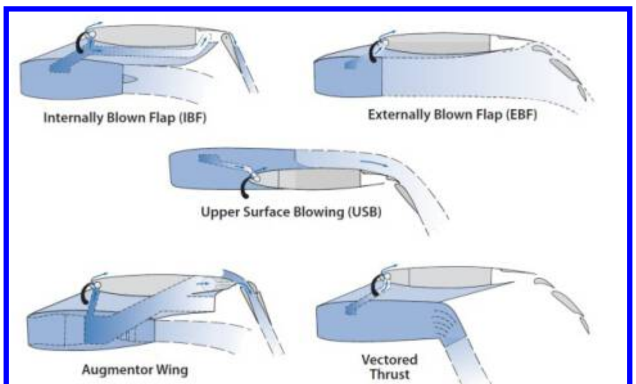

Figure 2. Various types of powered high lift devices that can be used with flaps. [5]

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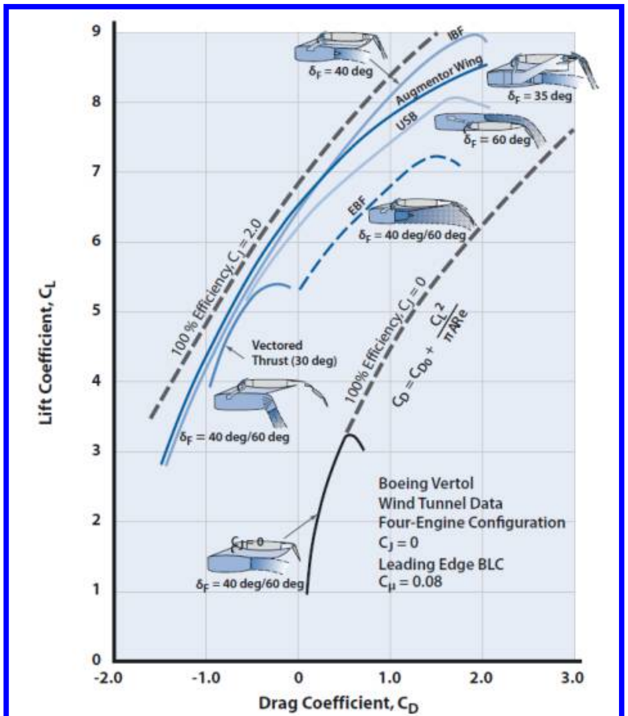

Figure 3. Lift and drag performance of various types of powered high lift devices using flaps. [5]

One modern example of organizations utilizing the aerodynamic benefits of blown wings is the NASA's X-57 Maxwell shown in Figure 4, the agency's first fully electric experimental aircraft. [8] Developed as a modification to the Tecnam P2006T, the Maxwell program is helping to learn more about the use of DEP. [9] Two large motors located at the wingtips are used for cruise flight while the smaller motors distributed across the wingspan provide additional thrust and lift during takeoff and landing. The high lift propellers blow air across the upper surface like the USB configuration but are centered slightly below the wing to also be used as an EBF for generating high levels of lift. These small propellers were unconventionally designed to maximize the exit air velocity as well as providing thrust because the main goal of these propellers assist in generation of thrust as well as lift. [10] The two major modifications to the baseline aircraft include the DEP system and the smaller, higher aspect ratio wing.

Figure 4. A model of the designed NASA Maxwell DEP aircraft. [8]

One important item to note for these powered high lift devices, including those utilized on the NASA Maxwell, is that the lift generated depends upon the momentum of air blown across the surface. This is sometimes represented by the thrust coefficient

$$c_T = \frac{T}{q_\infty A} \tag{3}$$

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where T is the thrust generated by an individual propulsor, q_∞ is the freestream dynamic pressure, and A is the propulsion system frontal area. [11] An example of how the lift coefficient changes for one wing-engine combination is visible in Figure 5, results of a blown airfoil tested at the University of Illinois. [9] As the throttle setting goes from 0% to 100%, the lift coefficient almost doubles. This greatly complicates the integration of propulsion and the wing because an adjustment to the ducted fan throttle setting will quickly cause a considerable change in aircraft handling. On the other hand, the pitching momentum coefficient, C_m, is related to the lift coefficient, depends upon the wing-propulsor configuration, and must be studied for each configuration considered. The plot in Figure 6 shows the results of computational fluid dynamics (CFD) testing for a transonic blown wing transport aircraft. [12] An increase in the thrust from 50% to 93% causes the lift coefficient to increase, but it also significantly lowers the pitching moment coefficient, affecting the stability of the aircraft.

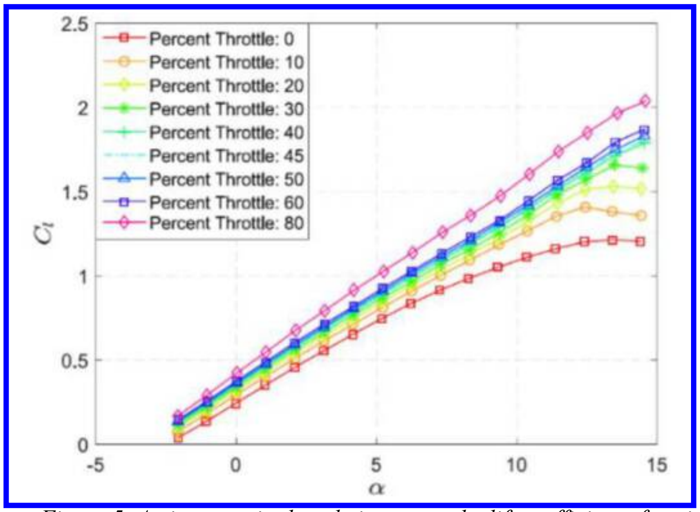

Figure 5. An increase in throttle increases the lift coefficient of a wing. [9]

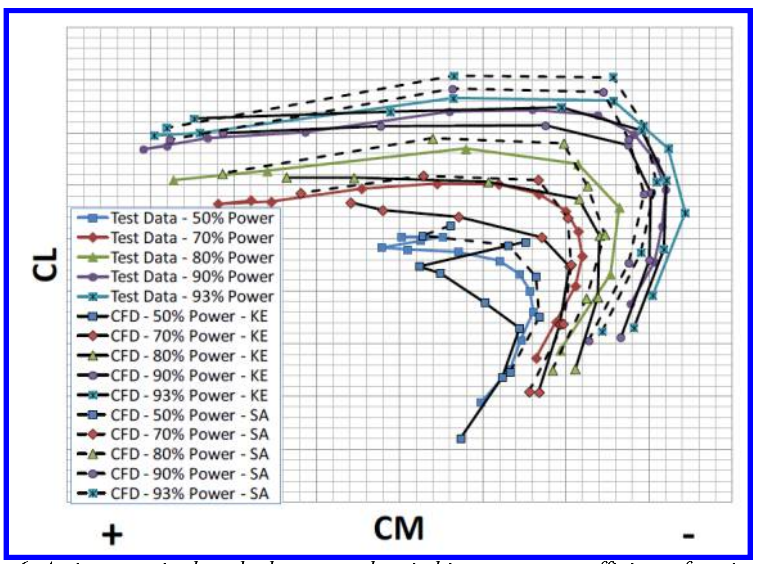

Figure 6. An increase in throttle decreases the pitching moment coefficient of a wing. [12]

As evidenced by Figure 2 and Figure 3, there are many geometric and aerodynamic variables that play a role in the amount of lift and pitching moment generated by a wing. Some of these variables include the airfoil thickness, airfoil chord, airfoil camber, angle of attack, Reynolds number, Mach number, and aspect ratio, amongst others. When studying a blown wing, these factors are all important in addition to the engine geometry, engine location, the thrust coefficient, and more. There are no equations or theoretical means of predicting the lift or pitching moment coefficients while including all these parameters. Performance optimization of blown wing configurations become harder to predict when the number of influential variable increases. A quick and robust way to predict aerodynamic coefficients of blown wings can reduce the optimization and testing process for both designers and experimentalists. This paper documents the several statistical and machine learning analysis tools and techniques used to predict lift and pitching moment coefficients from multiple blown wing configurations to aid the blown wing research community to bridge the gap between research and application. A brief overview of machine learning and its application in aerospace engineering is mentioned below.

In aerospace engineering, machine learning algorithms have been useful for a variety of applications. Researchers at the Argonne National Laboratory generated reduced-order models using deep neural networks as a faster alternative to CFD. [13] They use this model to study flow past a RAE2822 airfoil in transonic flow. With only 80 training points in the models, these researchers accurately modelled the flow over the airfoil at multiple angles of attack. Research presented by Omer and Maulik [14] discusses the use of artificial neural networks to study turbulent flows around a wind turbine. The generated model proved effective for reducing computational costs while maintaining effectiveness when applying the model to simulations outside the training dataset. A simplified model using proper orthogonal decomposition (POD) and a feedforward neural network together was compared with a high-fidelity solver with an unsteady flow application inside a combustor in Switzerland. [15] They showed that simplifications of models, when done properly, can still be robust and match the high-fidelity solutions well for faster solutions. Researchers in the United Kingdom used a large dataset of NACA 0012 airfoil wind tunnel testing and utilized a neural network approach to predict the noise generated by an airfoil. [16] Through the use of machine learning, it was as much as 80% better than previously developed noise prediction methods. [17] Machine learning has the ability to form complex decision boundaries with a large feature set (all the input variables that goes into the model) and helps in determining the relationship between the features and target variable which is the lift coefficient in this scenario. Data science based techniques help in analyzing relationship within the features.

When working with blown wings, there are many variables which are referred to as features in the field of machine learning that play an important role in the amount of lift generated by a wing. Some of these features include the airfoil shape, the thrust coefficient, the angle of attack, and the propulsion system configuration. Performance of designs become harder to predict when many variables affect the performance of blown wings. A fast way to predict lift and pitching moment coefficients can reduce the turnaround time for experimentalists and designers. As such, machine learning and other appropriate numerical analysis techniques are developed to utilize existing data to understand the relationship between the lift coefficient and the multitude of other variables to qualitatively and quantitatively understand the underlying physics.

IV. Data Collection and Analysis

A. Data Collection

To train the machine learning algorithm, a thorough literature search ranging from NACA reports in 1950s till most recent ones on blown wing was performed to gather quantitative data on the configuration type, geometry and the resultant aerodynamic coefficients, some of which are represented in Figure 7. [18-28] A resurgence of publications in recent years demonstrates the renewed interest in blown wings. In order to preserve the accuracy of the predictions, the three-dimensional wind tunnel testing data from the literature was given higher priority than any CFD type investigations. The models investigated with these studies include propellers, turbojets, and ducted fans as propulsion sources mounted at multiple locations around the wings and operated at varying thrust settings.

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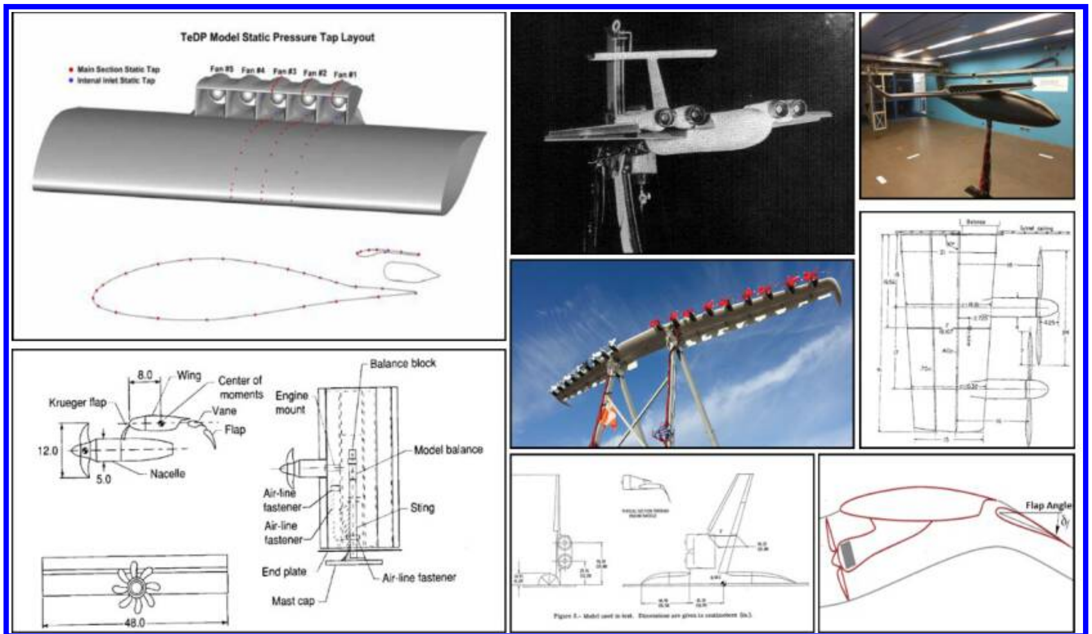

Figure 7. A variety of configurations and test conditions were studied during the data collection process. [18-25]

The references used for this study [18-28] provided a total of 705 data points for numerical analysis of the lift coefficient. Two studies did not report the pitching moment coefficient, so the dataset used to study the pitching moment coefficient had 645 variables. Table 1 below shows a summary of the tests described in these references. While some of the references also performed a CFD analysis of the same configuration, only wind tunnel data was retrieved from these reports as inputs to the prediction model. All experiments except for one varied the angle of attack (AoA) and thrust coefficient (C_T). Total number of data points accumulated from all databases is 705, varying greatly from 5 to 230 with an average contribution of 59 data points each. An experiment performed by Agrawal et al make up 33% of the dataset which likely skews the results to reflect those results more than the others.

Table 1. Various wind tunnel experiments were performed with unique setups and independent variables studied.

Reference	Propulsion System	Testing Performed	Independent Variables	Number of Data Points
Kerho (2015) [17]	Ducted Fan	Wind Tunnel	AoA, C_T	12
Phelps et al (1973) [18]	Turbofan	Wind Tunnel	AoA, C_T, Flap, Number of Motors	96
Dillinger et al (2018) [19]	Ducted Fan	Wind Tunnel	AoA, C_T	36
Deere (2017) [20]	Propeller	Wind Tunnel	AoA, C_T	21
Kuhn et al (1956) [21]	Propeller	Wind Tunnel	AoA, C_T	24
Gentry et al (1994) [22]	Propeller	Wind Tunnel	AoA, C_T, Flap, Chord Location	72
Phelps et al (1973) [23]	Turbofan	Wind Tunnel	AoA, C_T, Flap	36
Agrawal et al (2019) [24]	Propeller	Wind Tunnel	AoA, C_T, Flap	230
Langley (1976) [25]	Turbofan	Wind Tunnel	AoA, C_T, Flap	54
Aoyagi et al (1973) [26]	Turbofan	Wind Tunnel	AoA, C_T, Flap	48
Samuelsson (1987) [27]	Propeller	Wind Tunnel	AoA	5
Falarski et al (1971) [28]	Ducted Fan	Wind Tunnel	AoA, C_T, Flap	71

The variables collected from these documents for each data point are as follows:

- Wing Area, S
- Aspect Ratio, AR
- Wingspan, b
- Camber, C
- Thickness-to-chord ratio, t/c
- Flap Angle, δ
- Number of Motors, n
- Fan Height, h
- Fan Chord Location, x
- Fan Area, A
- Thrust Coefficient, T_C
- Angle of Attack, α
- Lift Coefficient, C_L
- Pitching Moment Coefficient, C_m

Another variable originally collected from these sources was the Reynolds number. This variable was ultimately left out of the analysis because it had a relatively low importance (to be discussed later) in comparison to the other variables. The other variables collectively captured wing geometry, propulsion geometry, and the propulsion thrust setting as defined in equation (3) used in the blown wing configuration. The Fan Height, h, is defined as the height of the propulsion system rotational center with respect to the wing leading edge, with positive sign denoting its location above the wing and vice-versa. Similarly, the Fan Chord Location, x, is defined as the distance between the propulsion system inlet/propellers and the leading edge, with positive sign denoting its location in front of the wing and vice-versa. These and several other geometric values measured during data collection are shown below in Figure 8. Both variables were nondimensionalized based on the chord length. The h and x were also used as qualitative variables. A chord location in front of the leading edge was labelled as 'Off Wing' while other locations were labelled as 'On Wing' to specify whether the propulsion system blew air over the entire wing or only a portion of it. Similarly, a positive 'Fan Height' value was labelled as 'Bottom Wing' while a negative value was labelled as 'Top Wing'. This helped to specify whether the propulsion system blowing was more dominant on the upper or lower surface. A five number summary for each variable collected is displayed in Table 2 below. This summary contains the minimum and maximum values of the dataset in addition to the first quartile (Q1), second quartile (the median) and third quartile (Q3).

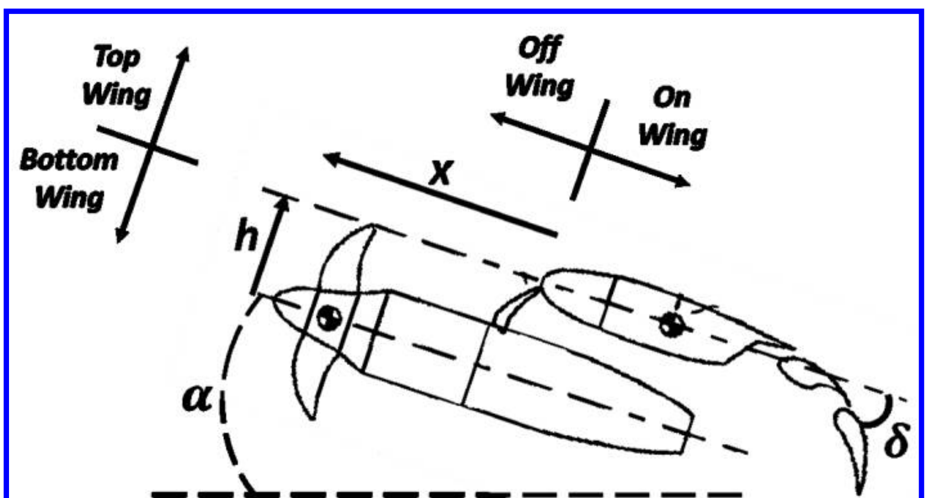

Figure 8. Several geometric features measured to generate the dataset.

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Table 2. The five number summary contains the minimum and maximum values of the dataset in addition to the quartiles.

Variable	Min	Q1	Median	Q3	Max
Wing Area, sq. ft.	1.5	1.5	11.2	159.5	212.5
Num of Motors	0	2	4	4	32
Fan Area, sq. ft.	0.02	0.18	0.18	2.66	12.57
Fan Chord Location	-0.95	-0.50	-0.06	0.25	0.90
Fan Height	-0.25	-0.24	-0.16	0.16	0.28
Thrust Coefficient	0.0	0.1	1.8	7.1	123.2
Airfoil Thickness	0.10	0.14	0.15	0.15	0.18
Camber	0.000	0.020	0.024	0.025	0.040
Flap Angle, deg.	0	20	40	55	90
Angle of Attack, deg.	0	2	4	8	14
Wingspan, ft.	2.0	2.0	6.6	29.0	57.3
Aspect Ratio	1.1	2.7	5.3	7.0	17.4
Lift Coefficient	-0.1	1.5	2.7	4.3	9.1
Pitching Coefficient	-4.9	-0.65	-0.35	-0.13	1.25

Finally, the following additional quantitative variables were defined based on the collected quantitative data:

$$T_q = T_C * A \tag{4}$$

$$\upsilon = t/c * C * h \tag{5}$$

$$R = \frac{A * n}{S} \tag{6}$$

$$WC = A * \frac{n}{b^2} \tag{7}$$

In the above equations, T_q is 'Thrust Per Density' (removing fan area as a contributing factor), v is 'Wing Volume', R is the fan 'Area Ratio', and WC is the 'Wing Coverage'. The addition of qualitative and quantitative variables to the original variables gathered from papers resulted in the final list of variables to be numerically analyzed, shown below in Table 3. These variables were used for modelling and numerical analysis.

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Table 3. The final list of variables used for modelling and numerical analysis.

Quantitative Variables	Qualitative Variables
Wing Area, sq. ft.	On Wing
Num of Motors	Off Wing
Fan Area, sq. ft.	Bottom Wing
Fan Height	Top Wing
Fan Chord Location	
Thrust Coefficient	
Thrust Per Pressure	
Volume	
Area Ratio	
Airfoil Thickness	
Camber	
Flap Angle, deg.	
Angle of Attack, deg.	
Wingspan, ft.	
Aspect Ratio	
Wing Coverage	
Lift Coefficient	
Pitching Coefficient	

B. Analysis Tools

Several analysis methods were applied to better understand the inherent relationships between the variables/features within the data which are described in this section. These methods include calculating the Pearson correlation, studying the variance of data, and generating models using machine learning techniques.

a) Pearson correlation:

The Pearson correlation technique calculates a correlation value between each pair of input values that correspond with a given lift or moment coefficient. A correlation close to positive one says that one variable has an almost perfect direct relationship with the other. A correlation close to negative one says that one variable has an almost perfect inverse relationship with the other. The assumptions required for a Pearson correlation include variable continuity with a normal distribution, linearity, and homoscedasticity. This analysis allows viewers to see if expected correlations exist – this will help tell whether the data used for modelling makes sense. Additionally, it can point out unobvious relationships within the dataset that can be useful for designers. Two other correlation techniques briefly considered for analysis include the Kendall and Spearman methods. While not all assumptions – such as a linear relationship between variables – could be assumed, the Pearson correlation was deemed most appropriate for this analysis.

b) Variance:

Next, the variance contribution of each input variable was studied to determine what kind of spread the dataset has. The variance Var_a of each variable vector A_i is defined as

$$Var_a = \frac{1}{705-1}\sum_{i=1}^{705}|A_i - \mu|^2 \tag{8}$$

where μ is the mean value of each variable vector. The variance contribution of each variable Var_{cont_a} is defined as

$$Var_{cont_a} = Var_a \left[\sum_{a=1}^{N} Var_a \right]^{-1} \quad (9)$$

The sum of contributions for all N variables always equals unity. The variance contributions for each variable are all equal for ideal statistical analysis. The values in the dataset are taken from mostly independent reports and papers, so there is no control over what variance exists for the variables.

B.1 Prediction models

The data was also used for training a variety of prediction models which are briefly explained below. These models include the linear regression model, stepwise linear regression model, neural network (NN), support vector machine (SVM), random forest, and the Gaussian progression.

a) Linear regression:

Linear regression is a statistical approach that helps in describing a continuous variable as a function of one or more features. This type of regression approach helps in analyzing the relationship in complex systems. In addition, weights are assigned to each feature based on the training data that is used, thereby helping us understand the relationship between the target variable and features present in the dataset.

b) Stepwise linear regression

Stepwise linear regression methods have the ability to add or remove features depending on its performance starting from a constant model. It's a systematic approach to form a linear or generalized linear model based on their statistical significance in explaining the target variable (lift or moment coefficient). It is initialized with specific linear model, and later features are added or removed depending on its performance. Statistical significance is determined based on the p-value statistic.

c) Neural networks

Neural networks have been adopted in several applications ranging from continuous number based, computer vision, and natural language processing applications. Neural networks have the ability to determine the complex relationships that exists between the features and target values. Several layers and nodes are added to determine such complicated relationships. In this paper, the performance is analyzed when using a single hidden layer with 100 nodes.

d) Support Vector Machine

Support Vector Machine (SVM) analysis is a popular machine learning tool used for both classification and regression approaches. It's a non-parametric method due to its reliance on kernel functions. SVM helps in minimizing the error margin and in finding a function that deviates the target value by a value no greater than the given threshold in an optimal hyperplane. In this research, a kernel is used with a polynomial order of three.

e) Random Forest

Random forest is a regression tree ensemble method that is composed of weighted combination of multiple regression trees. Theoretically, combining multiple regression methods helps in maximizing performance and minimizing error. Trees are determined based on the branch of features that help in sub-dividing the dataset that would help in predicting the target value.

f) Gaussian Progression

Gaussian progression method is a non-parametric kernel based probabilistic approach. It helps in determining the prediction intervals using the trained model. Gaussian progression regression model's equation is similar to linear regression. Covariance function of the latent variables helps in capturing the smoothness of the response and functions that can help in projecting the input features for minimal error.

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B.2 Data Training

Before training models, the data for each variable was normalized using Equation 10

$$S_{a_{norm}} = \frac{S_a - \bar{S}}{(S_{max} - S_{min})} \tag{10}$$

where S_a is each data point in the vector of data with each variable, $\bar{S}$ is the average value, S_{max} is the maximum value for each variable, and S_{min} is the minimum value for each variable. The maximum and minimum can be seen in Table 2. For each variable, the values associated with that variable in the entire dataset ranged from negative one to positive one. The data was normalized to prevent skewed results due to division or multiplication of large numbers (such as Reynolds numbers) and small numbers (such as camber) in the dataset. Next, 80% of the data was randomly selected to be used for generating models while 20% was set aside as a validation dataset. These models were then compared with the actual lift or pitching moment coefficient values within the validation dataset to determine the median percent error of each model. The lower the percent error, the better a model can predict the validation data set lift or moment coefficients based on input variables.

a) 10-fold cross validation

A 10-fold cross validation was later performed to generate a more robust set of models. When studying the performance of 10-fold cross validation where the dataset is split into 10-folds, testing is implemented on one fold while the remaining folds are used for training and this is cycled over the entire suite.

b) Sequential Forward Selection (SFS)

Variable selection methods were used to calculate the relative importance of each variable for predicting the two coefficients. Sequential forward selection (SFS) was first used with each of the machine learning models to generate multiple rankings that could be compared. To perform SFS, each of the models were generated with only one input variable in the model. The variable that resulted in the lowest percent error value for lift or moment coefficient was listed as the highest importance. More variables were added to each of the models to improve the prediction method and theoretically reduce error. This type of selection is utilized to avoid exhaustive enumeration. Note that only the training dataset is implemented for this study. Figure 9 shows how the percent error could change with the addition of variables to the model. Eventually, more input variables would no longer improve the error, after which variables were no longer added to the model. The error value for each model reduced at varying rates, and the variable order (the ranking) varied from model to model.

B.3 Feature Selection Methods

The following feature selection methods were also investigated: minimum redundancy maximum relevance (MRMR), neighborhood component analysis (NCA), relief, and permutation. These provided additional variable importance rankings to compare. These selection methods are briefly described below.

a) MRMR algorithm

MRMR algorithm ranks the features and finds an optimal suite that are mutually and maximally dissimilar which can represent the target variable effectively. This algorithm helps in minimizing redundancy of feature set and maximizes the relevance to the feature set.

b) NCA

NCA helps in performing feature selection by assigning the feature weights by a diagonal adaptation of neighborhood component analysis with regularization.

c) Relief

Relief method helps in ranking features based on a set of *k* nearest neighbors based on the target variable.

d) Permutation

Permutation importance helps in ranking features by randomly changing the values of a given feature and how adverse it affects the performance of the regression algorithm.

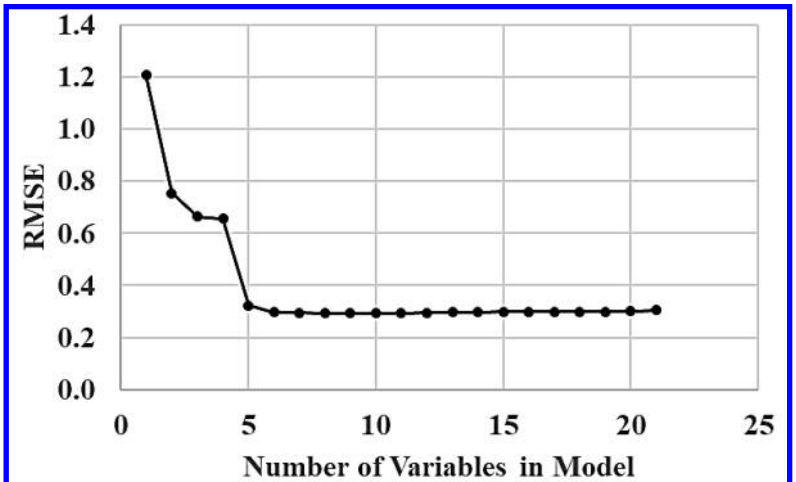

Figure 9. As input variables are added to the model, it can better predict the lift coefficient.

V. Results

A. Two-Dimensional Plots

Several two-dimensional plots of the data are shown in Figure 10 to indicate the general trends seen in the dataset. Figure 10 (a) seems to show a general trend of an increase in the lift coefficient with angle of attack, but its dependency on other variables such as engine placement and height are not discernible. Similarly, Figure 10 (b) seems to point to an ideal flap angle for maximizing the lift coefficient about 45 or 50 degrees but little else can be seen. Other plots of the collected data, like in Figure 10 (c) and (d), provide little useful information because the data is not well distributed or has large holes in it. These plots have limited capabilities for analyzing the data because there are too many other variables changing at once that cannot be captured.

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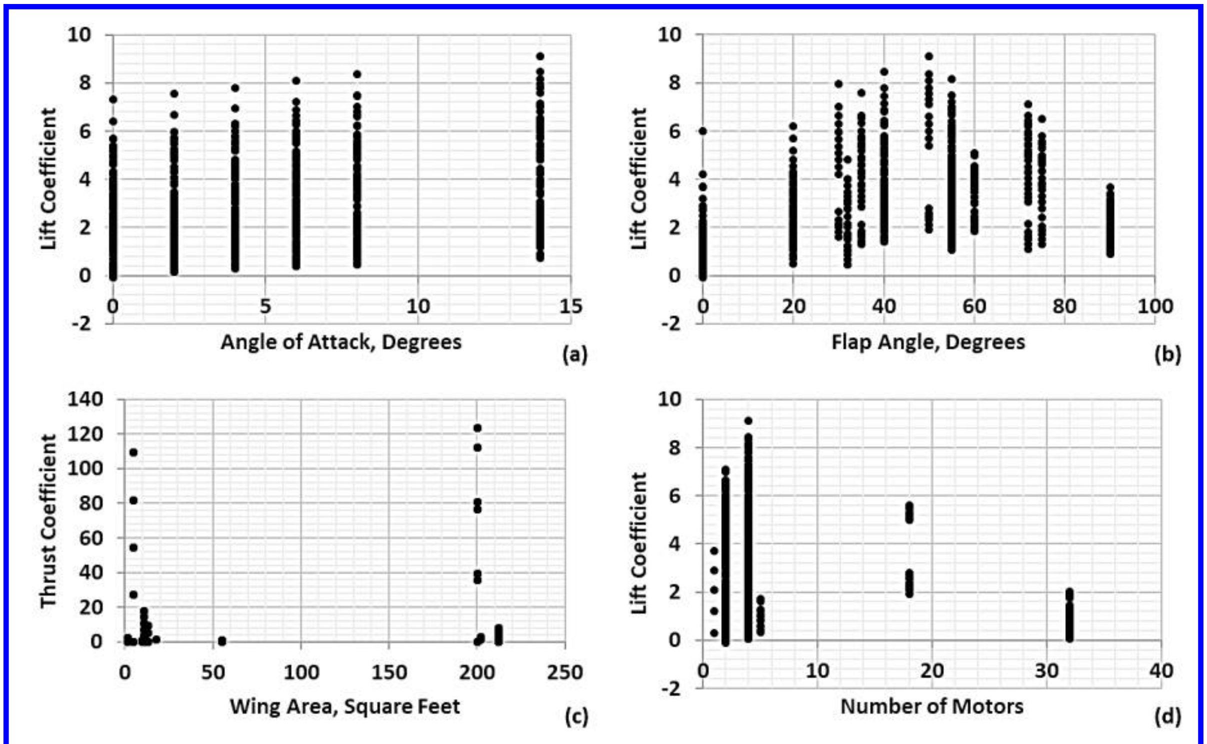

Figure 10. Two dimensional plots have limited capabilities for showing trends in data that have many input variables.

B. Pearson Correlation

The results from Pearson correlations are shown in Figure 11. This result is only shown for the lift coefficient because the pitching moment coefficient dataset is overall very similar and had agreeable results. One of the strongest correlations was between 'Wing Coverage' and 'Area Ratio'. This relationship is strong because large engines will cover more of the wingspan and therefore increase both values. Seeing such obvious and physical correlations like this in the data provides a pseudo validation to the Pearson correlation analysis and gives confidence in the correlation seen in between other variables. Expected relationships appearing in the Pearson correlation also support trust in the following analysis of the data. A comparatively non-obvious negative correlation is seen between 'Fan Area' and 'Camber'. As an attempt to explain this correlation from a physical perspective, a large propulsion system generally indicates a larger and faster aircraft and vice-versa. Aircraft with high Reynolds numbers usually have wings with low camber to minimize zero-lift drag coefficient. Another non-obvious strong correlation was discovered between 'Wing Area' and 'Fan Height'. Increasing the wing area of an aircraft increases the Reynolds number and is usually used at higher speeds. This negative correlation indicates that on aircraft designed to operate at high speeds (with a larger Reynolds number), the propulsion system is usually installed below the wing and not above. This is corroborated by a distributed propulsion CFD analysis performed by Wick [29] which found the ideal position for ducted fans on a transonic transport is below the wing at the trailing edge.

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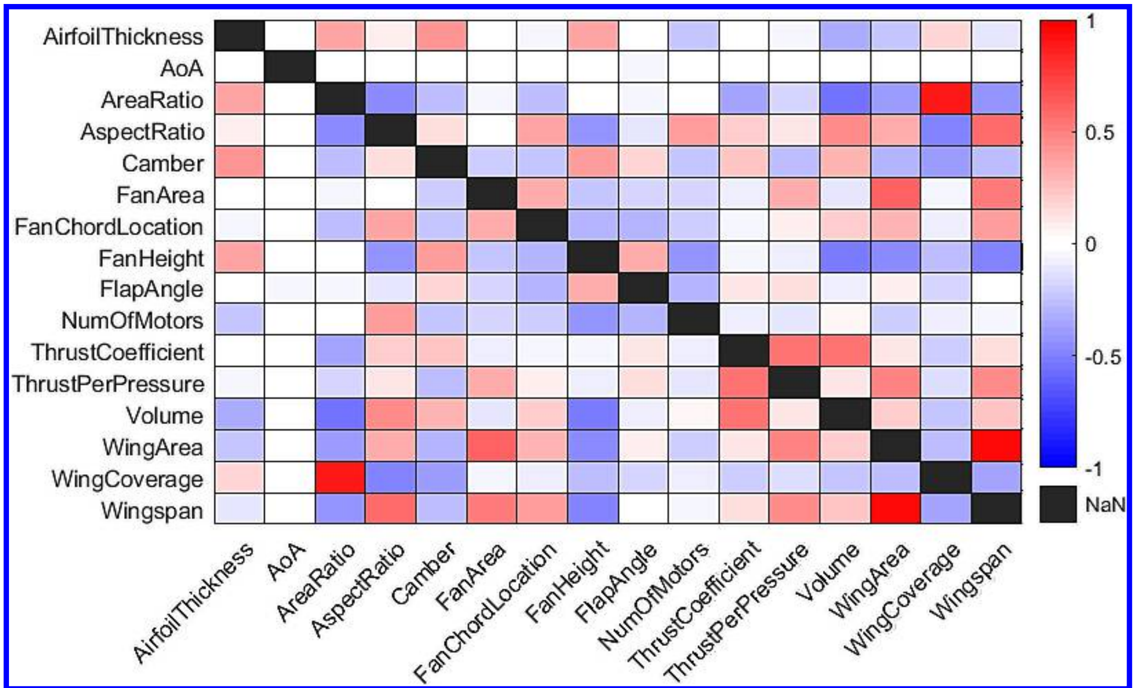

Figure 11. A Pearson correlation of the variables to see how each are connected.

C. Variance

As previously mentioned, an ideal case would be to have each of these variables contribute equally to the overall variance of the dataset. However, as seen in Figure 12, this is not the case. The parameter with highest variance in the dataset is the Wing Area and the parameters with the least variance are the airfoil thickness and the thrust per pressure. Variables with low variance indicate the lack of a wide range of data specific to that variable. For example, the Number of Motors category has one of the lowest variances as visualized in Figure 10 (d). This is because most of the blown wing investigations were performed with either 0 or 5 motors with few exceptions. Parametric investigations on blown wings must be performed to help improve the individual contributions of each variable, especially the ones which are on the right side of Figure 12.

D. Coefficient Predictions

When generating these models, the predicted coefficients of the validation dataset were compared with the true values pulled from the raw dataset. As an example, the predicted lift coefficient results from the Gaussian progression model are shown in Figure 13 and compared with the actual values for each data point. The pitching moment coefficient results are shown in Figure 14. Recall that the "actual values" indicated in Figure 13 and Figure 14 are the 20% of the dataset which have been randomly selected. This dataset is not included to train the model and only serves as a validation dataset. Since the validation dataset was randomly generated, the aerodynamic coefficient pairs from each data point are randomly scattered throughout the plot, except for some outliers in the pitching moment coefficient data. It is visually clear that the predictions closely match the true values for both coefficients, including prediction of the outliers in the pitching moment coefficient.

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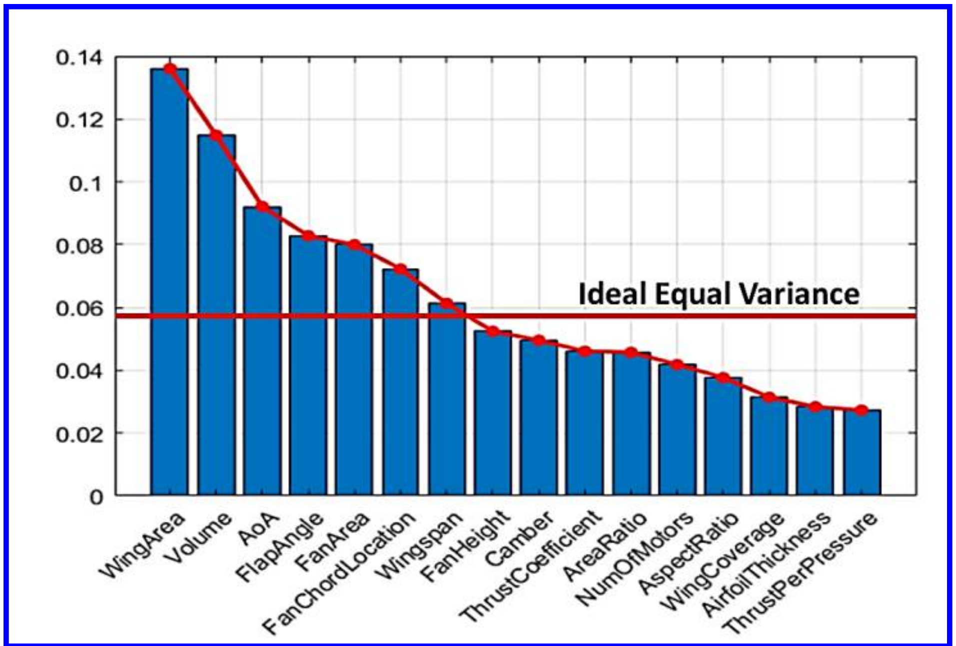

Figure 12. The variance of each variable used in modelling; there is a wide range of variance values.

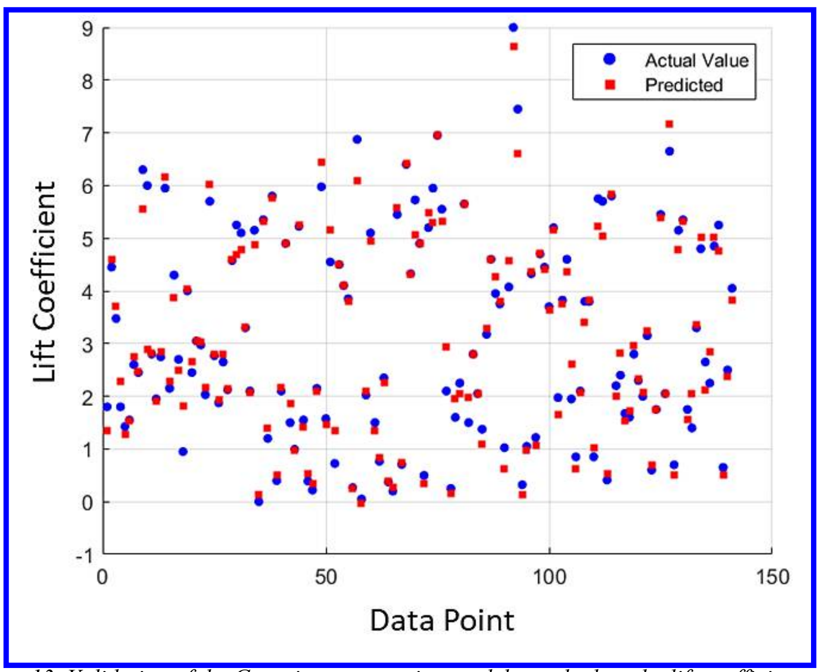

Figure 13. Validation of the Gaussian progression model to calculate the lift coefficient error.

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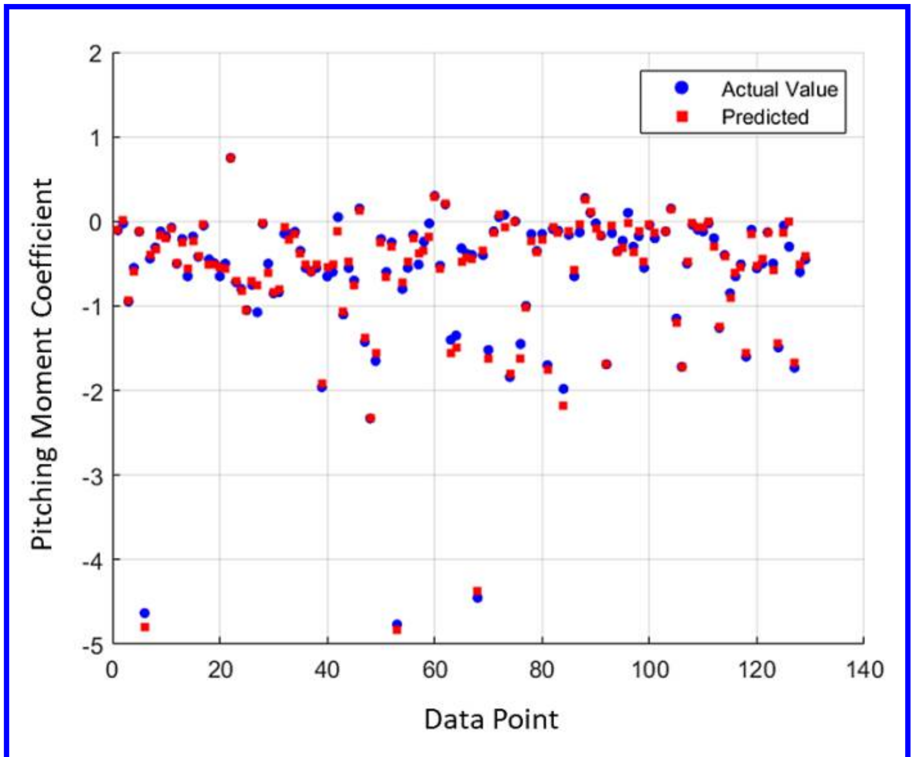

Figure 14. Validation of the Gaussian progression model to calculate the pitching moment coefficient error.

Figure 15 shows a histogram of the error from the scatter plots generated from each model for the lift coefficient. The pitching moment coefficient results are displayed in Figure 16. The zero error point is shown with a vertical bar in the center of each plot. A percent error value for each model is displayed above each of the subplots. The two worst performing models for both coefficients, those with the largest error, were the linear model and stepwise linear model. The linear models likely have a high error because not all input variables have a linear relationship with the aerodynamic coefficients. The best two performing models for both coefficients are the NN and Gaussian progression models. These models can predict the lift coefficient of an engine/wing configuration with an accuracy of roughly 4% (8.0-9.5% for moment coefficient). This can be extremely useful when performing preliminary aircraft design space exploration or estimating forces and moments to be expected for wind tunnel testing or CFD.

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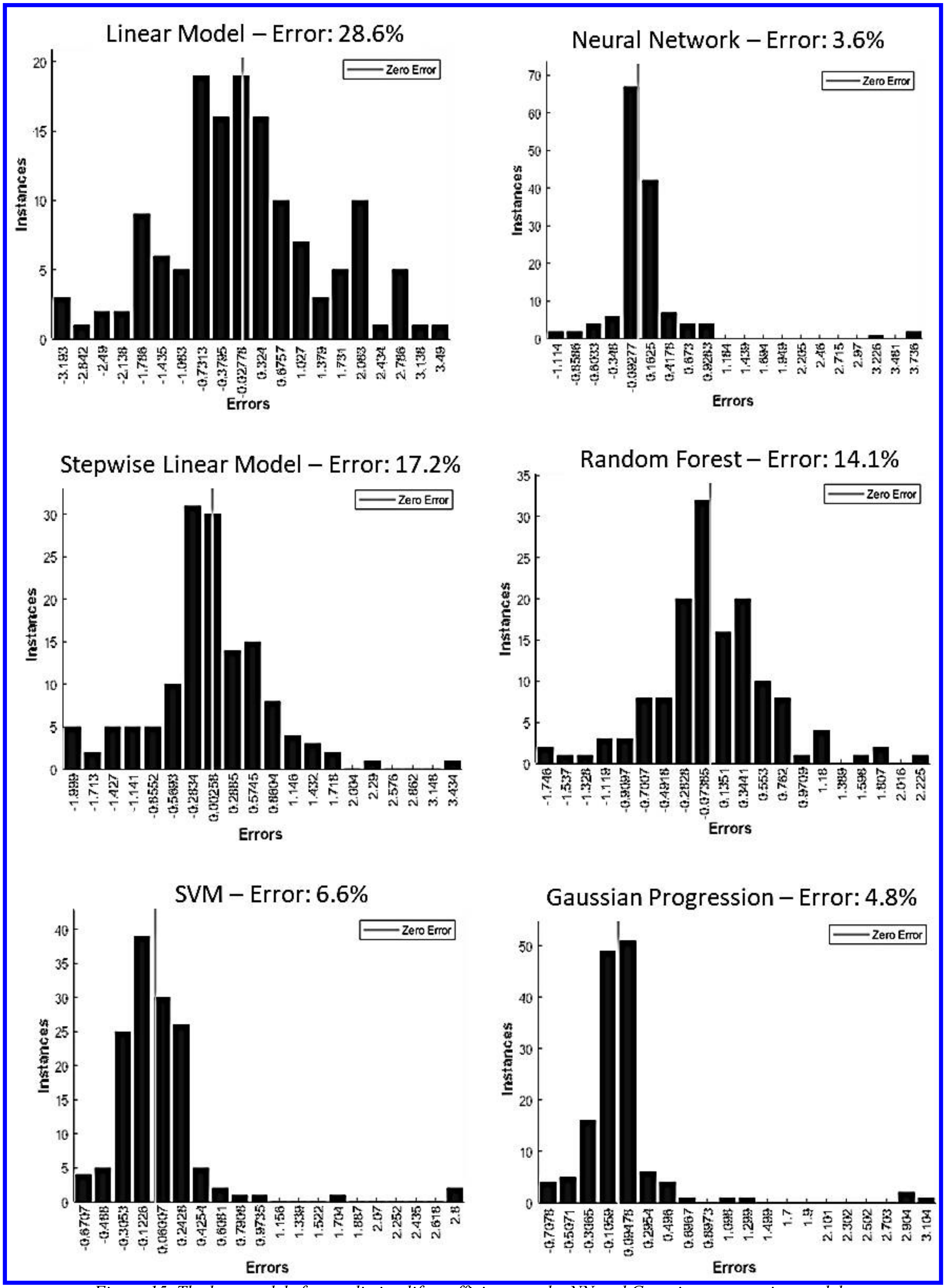

Figure 15. The best models for predicting lift coefficient are the NN and Gaussian progression models.

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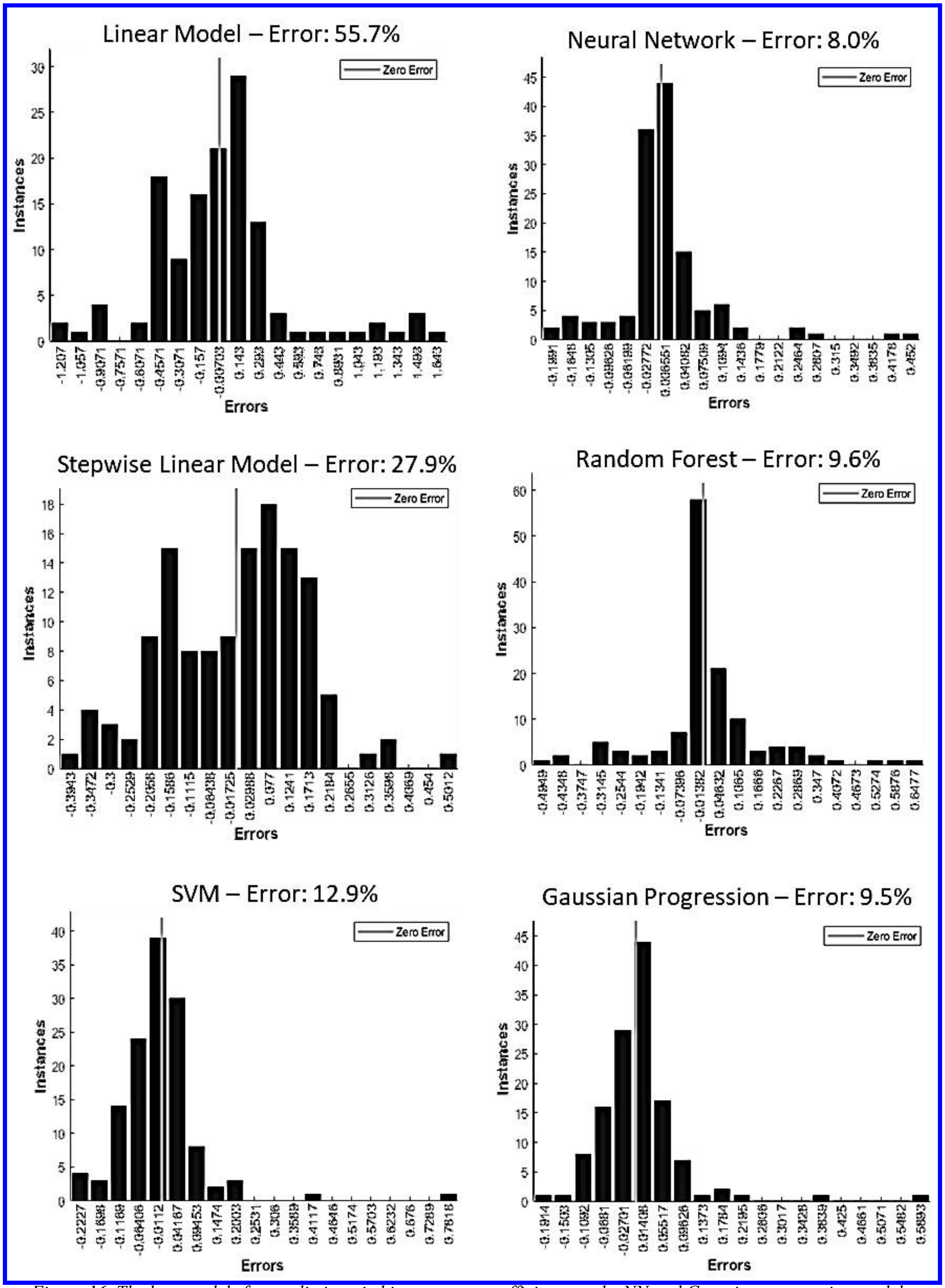

Figure 16. The best models for predicting pitching moment coefficient are the NN and Gaussian progression models.

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E. Determination of Variable Importance

Results of the 10-fold cross validation are shown in Table 4 for the lift coefficient and Table 5 for the pitching moment coefficient. The first column lists the percent error values of each model before cross validation, repeated from Figure 15 and Figure 16. The second column in each table lists the updated results after performing the 10-fold cross validation. The third column lists results of a 10-fold cross validation with fewer input variables for the model than the original 20. Selection of the number of features to include for each model were based on the results like those shown in Figure 9. The number of features were selected based on how many features were required to minimize the percent error. The Gaussian model required the fewest features (5) for the lift coefficient prediction, and it still had the lowest error. These features, listed high to low in order of importance, are: 'Thrust Coefficient', 'Flap Angle', 'Area Ratio', 'Angle of Attack', and 'Fan Height'. If a black box type device were to be generated that output a lift coefficient estimation with multiple input variables for a blown wing, the Gaussian method can consistently provide an estimation within about 2.5% of the true value. As for the pitching moment coefficient, the Gaussian model still has the lowest error, but 13 of the 20 features were required to do that. That being said, an error of 3.8% can be achieved with the Gaussian model by including 10 features. When using 10 features, the NN has an error of 6.7%. When setting up a black box device to predict the pitching moment coefficient, designers would need to balance accuracy and the number of required input variables.

Table 4. NN and the Gaussian models consistently are the best predictors of the lift coefficient with SVM as a close third.

	Before 10-fold Validation	After 10-fold Validation	10-fold Validation with Less Features	
Classification Model	Error (%)	Error (%)	Number of Features	Error (%)
Stepwise	17.2	19.4	13/20	14.1
NN	3.6	4.2	10/20	3.2
SVM	6.6	6.1	12/20	5.5
Linear	28.6	32.1	16/20	29.5
Tree	14.1	13.6	9/20	12.5
Gaussian	4.8	4.3	5/20	2.5

Table 5. NN and the Gaussian models consistently are the best predictors of the moment coefficient.

	Before 10-fold Validation	After 10-fold Validation	10-fold Validation with Less Features	
Classification Model	Error (%)	Error (%)	Number of Features	Error (%)
Stepwise	27.9	29.5	13/20	28
NN	8.0	7.9	16/20	5.1
SVM	12.9	9.4	11/20	7.6
Linear	55.7	47.9	13/20	42.6
Tree	9.6	9.8	3/20	9.1
Gaussian	9.5	7.8	13/20	3.5

The selection of top features to use with the 10-fold cross validation relied on the lift coefficient results of the SFS study shown in Table 6. Each column lists the first seven variables used when generating plots like the one in Figure 9 that are considered here as the most important drivers of the lift coefficient for a blown wing. Results of the same study performed to study the pitching moment coefficient are shown in Table 7. 'Thrust Coefficient' is the highest ranking variable for all models except SVM and NN (NN only differs for the moment coefficient). The only other variable to be in the top position for any model of the lift coefficient is 'Flap Angle'. Both variables were expected to rank high because each are known to greatly increase the lift coefficient. For example, the lift coefficient curves in Figure 5 are significantly shifted upwards due to an increase in the thrust generated. The SLM and linear model listing 'Wing Coverage' as the highest ranked variable for the pitching coefficient show that the contributing variables are not as clear as with the lift coefficient. After the top two ranks, the models for both coefficients begin to disagree on which variables are highly important. Additional rankings for lift and pitching moment coefficients generated via different modelling techniques are shown in Table 8 and Table 9 for comparison, most of which agree that 'Thrust Coefficient' and 'Flap Angle' are highly important variables. None of the additional modelling techniques listed 'Wing Coverage' as an important variable for predicting the pitching moment coefficient.

Table 6. The two clear large contributors to the lift coefficient are the thrust coefficient and flap angle.

Ranking	Linear Model	Gaussian Progression	Neural Network	SVM Model	Tree Regression	SLM Model
1	Thrust Coefficient	Thrust Coefficient	Thrust Coefficient	Flap Angle	Thrust Coefficient	Thrust Coefficient
2	Num Of Motors	Flap Angle	Flap Angle	AoA	Flap Angle	Wing Coverage
3	Off Wing	Area Ratio	Aspect Ratio	Volume	Wing Coverage	AoA
4	AoA	AoA	Wing Coverage	Thrust Coefficient	AoA	Num Of Motors
5	Airfoil Thickness	Fan Height	AoA	Area Ratio	Fan Height	Airfoil Thickness
6	Volume	Wingspan	Num Of Motors	Fan Height	Fan Chord Location	Thrust Per Pressure
7	Wing Coverage	Num Of Motors	Camber	Off Wing	Volume	Area Ratio

Table 7. The models did not agree on top contributors to the pitching moment coefficient like they did with lift.

Ranking	Linear Model	Gaussian Progression	Neural Network	SVM Model	Tree Regression	SLM Model
1	Wing Coverage	Thrust Coefficient	Flap Angle	Flap Angle	Thrust Coefficient	Wing Coverage
2	Area Ratio	Flap Angle	Thrust Coefficient	Aspect Ratio	Flap Angle	Aspect Ratio
3	Thrust Coefficient	Wing Coverage	Wing Area	Wing Coverage	Wing Coverage	Fan Area
4	Thrust Per Pressure	Bottom Wing	Top Wing	On Wing	Wing Area	Camber
5	Volume	Top Wing	Area Ratio	Bottom Wing	Fan Area	Thrust Per Pressure
6	On Wing	Area Ratio	Wing Coverage	AoA	Volume	Airfoil Thickness
7	Fan Area	Wingspan	Bottom Wing	Thrust Coefficient	Off Wing	Flap Angle

Table 8. The additional ranking methods disagreed in what variables have large contributions to the lift coefficient.

Ranking	MRMR	NCA	Relief	Permutation
1	Flap Angle	Thrust Coefficient	Flap Angle	Flap Angle
2	AoA	Num Of Motors	Thrust Coefficient	Thrust Per Pressure
3	Thrust Per Pressure	Thrust Per Pressure	Thrust Per Pressure	Thrust Coefficient
4	Num Of Motors	Flap Angle	Camber	AoA
5	Fan Height	Area Ratio	Fan Height	Aspect Ratio
6	Thrust Coefficient	AoA	Num Of Motors	Wingspan
7	Fan Area	Camber	Aspect Ratio	Area Ratio

Table 9. The additional ranking methods disagree in what variables have large contributions to the pitching coefficient.

Ranking	MRMR	NCA	Relief	Permutation
1	Thrust Coefficient	Thrust Coefficient	Flap Angle	Area Ratio
2	AoA	Flap Angle	Thrust Coefficient	Thrust Coefficient
3	Flap Angle	Aspect Ratio	Thrust Per Pressure	Flap Angle
4	Num Of Motors	Thrust Per Pressure	Camber	Fan Area
5	Aspect Ratio	Fan Height	Fan Height	Wingspan
6	Thrust Per Pressure	Airfoil Thickness	Wing Coverage	Volume
7	Camber	Wingspan	Aspect Ratio	AoA

To determine an overall ranking of the lift and pitching coefficients across the nine ranking methods used in Table 6 through Table 9, the average rank of each variable was calculated. The top seven average rankings with their ranking standard deviation are listed in Table 10 for the lift coefficient and Table 11 for the pitching moment coefficient. As expected, 'Thrust Coefficient' and 'Flap Angle' were the two most important variables for determining the lift and pitching coefficients, and the 'Thrust Coefficient' low standard deviation shows that the models agreed. The standard deviation becomes large after the top ranked variable, so there is much less agreement between the nine models. Comparing these top ranked variables with the variance in Figure 12, five of the seven most important input variables (the five variables differ for the pitching moment and lift coefficient rankings) have a variance contribution

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below the ideal variance level, including the variable with the lowest variance contribution (for the pitching moment coefficient). If these input variables indeed make a large contribution to determining the output lift coefficient, these variables should be the focus of future blown wing research.

Table 10. Thrust coefficient and flap angle were the highest ranking, but flap angle had a much higher standard deviation.

Parameter	Average Rank	Standard Deviation
Thrust Coefficient	2.1	1.64
Flap Angle	3.8	4.56
AoA	5.4	5.00
Num Of Motors	7.2	5.17
Area Ratio	8.1	3.14
Fan Height	8.8	4.14
Wing Coverage	8.8	4.40

Table 11. The thrust coefficient is the most important variable for both the lift and pitching moment coefficients.

Parameter	Average Rank	Standard Deviation
Thrust Coefficient	2.8	2.44
Flap Angle	3.4	3.32
Wing Coverage	5.5	3.98
Aspect Ratio	7	3.71
Area Ratio	7.8	4.07
Thrust Per Pressure	8.7	5.27
Fan Area	9	4.31

VI. Conclusions and Future Work

Machine learning techniques allowed for published data on blown wing research to be used for generating prediction models for lift and pitching moment coefficients given a variety of input variables. The Gaussian progression model proved to be the most effective model for predicting both coefficients because it had the least error when compared to validation datasets and required the least number of input variables to make an accurate prediction. Multiple methods were also used to rank the relative importance of each input variable. The results of this ranking show that the ‘Thrust Coefficient’ and ‘Flap Angle’ are the most important variables for predicting the two coefficients. More data is needed on the variables with lower variance, especially those that have a high importance to the lift and pitching coefficients. If additional data with greater variance is added to the dataset for the important variables, their importance can be confirmed or adjusted. In the future, additional collected data will be added to the dataset to improve modelling and ranking results. Additionally, clustering will be performed to display what input variable value ranges are appropriate when aiming for a specific lift and pitching moment coefficient output.

VII. Acknowledgements

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